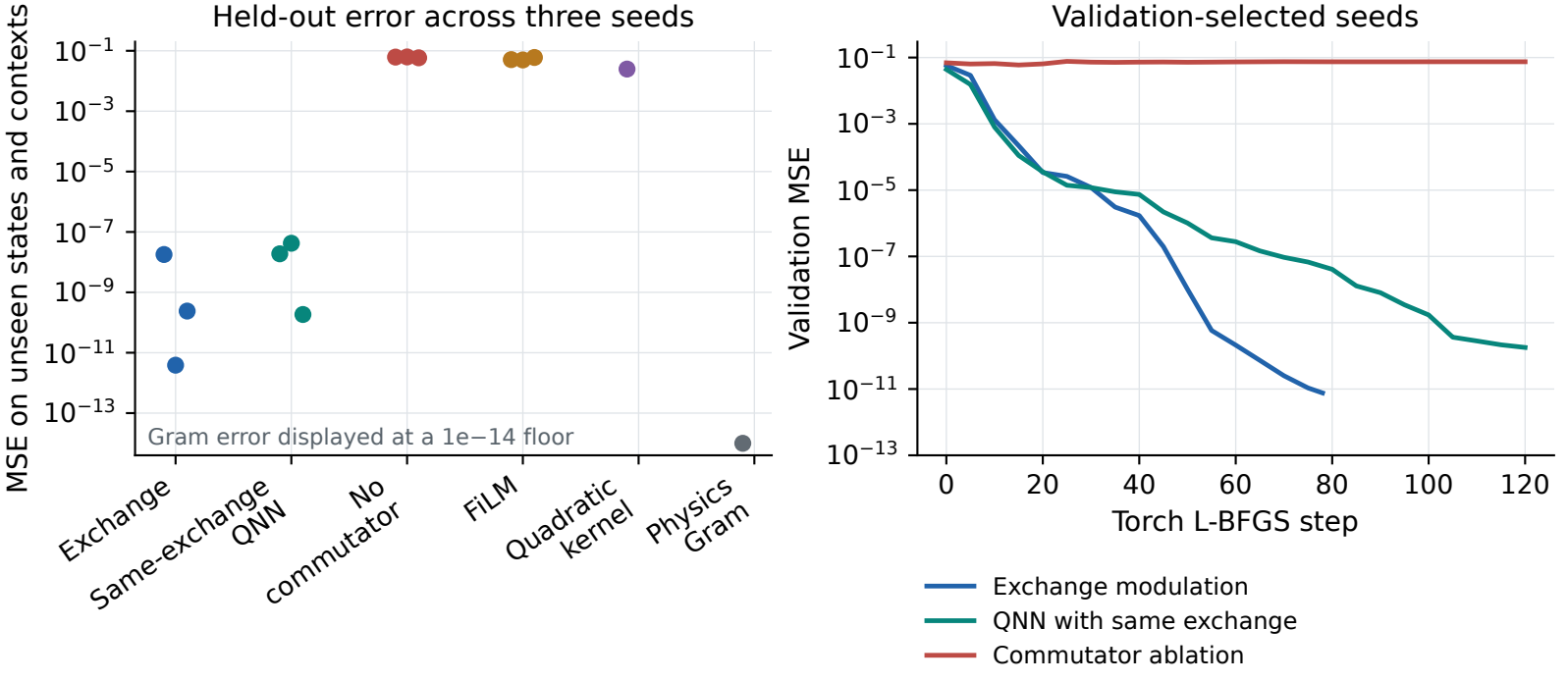
Held-out error across three seeds
MSE on unseen states and contexts
$10^{-1}$
$10^{-3}$
$10^{-5}$
$10^{-7}$
$10^{-9}$
$10^{-11}$
$10^{-13}$
Gram error displayed at a 1e−14 floor
Exchange
Same-exchange QNN
No commutator
FiLM
Quadratic kernel
Physics Gram
Validation-selected seeds
Validation MSE
$10^{-1}$
$10^{-3}$
$10^{-5}$
$10^{-7}$
$10^{-9}$
$10^{-11}$
$10^{-13}$
0
20
40
60
80
100
120
Torch L-BFGS step
Exchange modulation
QNN with same exchange
Commutator ablation